

Assignment 1:

Linux File System & Permissions Scenario

Scenario:

You are a new DevOps engineer tasked with setting up a shared project directory on a Linux server for a team of developers. The directory `/opt/project_alpha` needs to be accessible by all members of the devteam group, but not by other users on the system. Additionally, all new files created within this directory by devteam members should automatically inherit the devteam group ownership.

Tasks:

1. Create a new group named devteam .
2. Create three new users: alice , bob , and devteam group.
3. Create the directory charlie . Add all three users to the `/opt/project_alpha` .
4. Set the ownership of `/opt/project_alpha` to root:devteam .
5. Configure the permissions for `/opt/project_alpha` such that: The owner (root) has full read, write, and execute permissions. The group (devteam) has read, write, and execute permissions. Others have no permissions.
6. Ensure that any new files or subdirectories created within `/opt/project_alpha` automatically inherit the devteam group ownership (using the SetGID bit).

7. Verification:

Log in as alice and create a file Verify its group ownership. Log in as a user not in test_alice.txt in `/opt/project_alpha` . devteam (e.g., guestuser , if available, or create one) and attempt to create a file in `/opt/project_alpha` . Document the outcome.

Expected Output:

- Commands used for each step.
- Output of `ls -ld /opt/project_alpha` after permissions are set.
- Output of `ls -l /opt/project_alpha` after alice creates a file.
- Error message (if any) when a non-devteam user attempts to create a file.

Create the devteam group

\$: sudo groupadd devteam

Verify:

\$: getent group devteam

Create users alice, bob, and Charlie

\$: sudo useradd -m alice

\$: sudo useradd -m bob

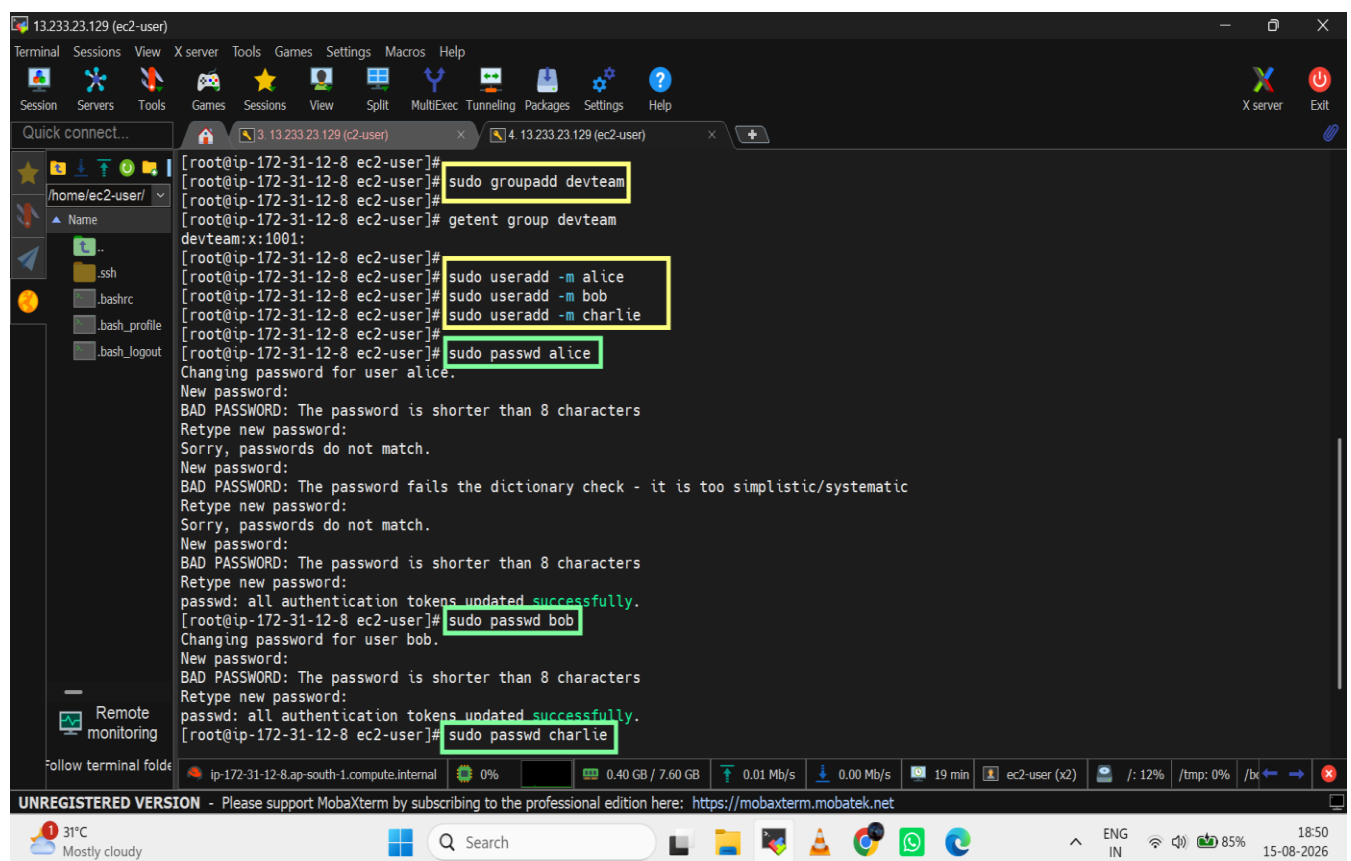
\$: sudo useradd -m charlie

Set passwords:

\$: sudo passwd alice

\$: sudo passwd bob

\$: sudo passwd charlie



The screenshot shows a MobaXterm terminal window with the following commands and output:

```
[root@ip-172-31-12-8 ec2-user]# sudo groupadd devteam
[root@ip-172-31-12-8 ec2-user]# getent group devteam
devteam:x:1001:
[root@ip-172-31-12-8 ec2-user]# sudo useradd -m alice
[root@ip-172-31-12-8 ec2-user]# sudo useradd -m bob
[root@ip-172-31-12-8 ec2-user]# sudo useradd -m charlie
[root@ip-172-31-12-8 ec2-user]# sudo passwd alice
Changing password for user alice.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-12-8 ec2-user]# sudo passwd bob
Changing password for user bob.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-12-8 ec2-user]# sudo passwd charlie
```

The terminal window also shows a sidebar with file explorer, a top menu bar, and a status bar at the bottom with system information like temperature, search, and battery level.

Add all three users to devteam

\$: sudo usermod -aG devteam alice

\$: sudo usermod -aG devteam bob

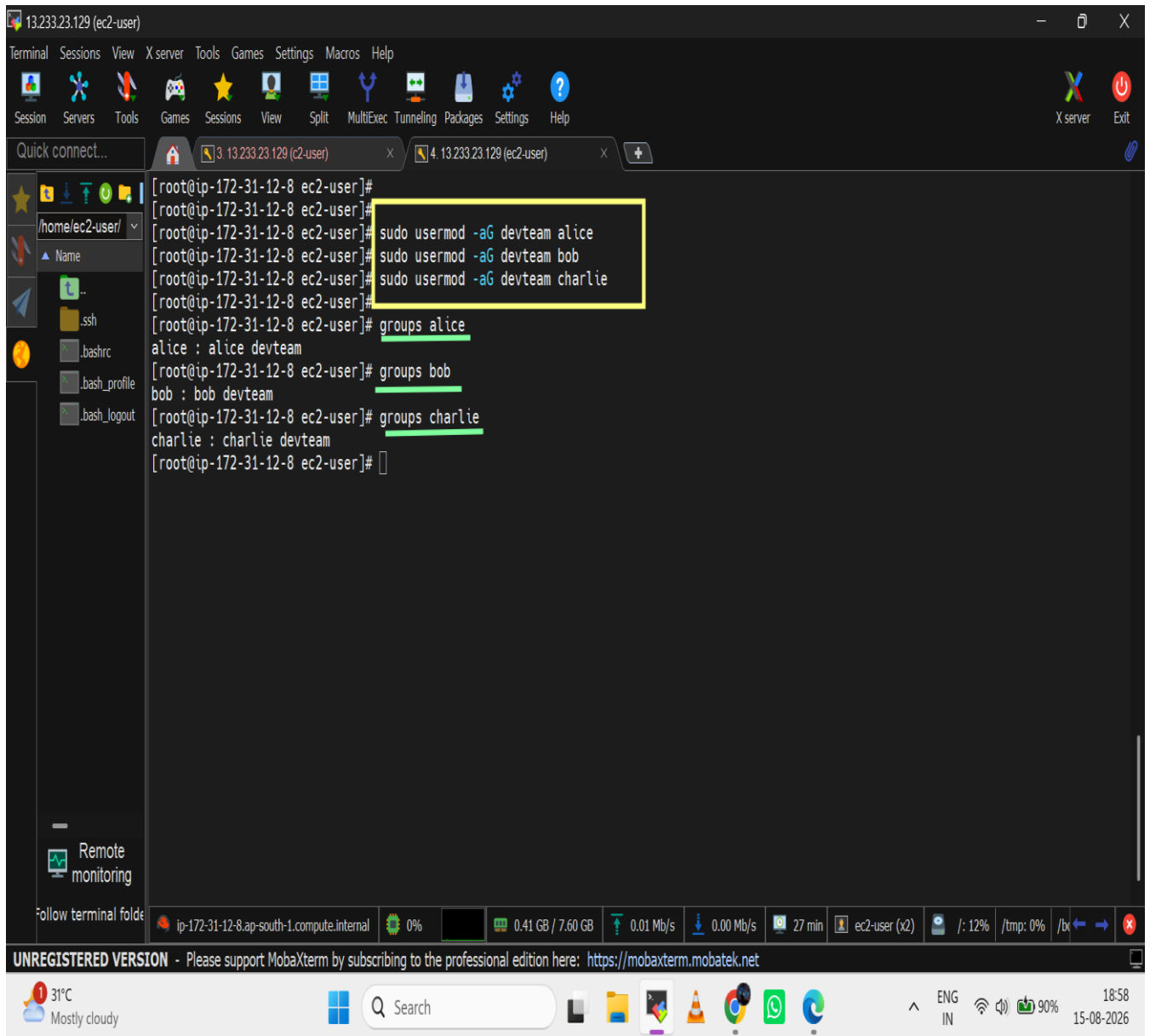
\$: sudo usermod -aG devteam Charlie

Verify:

\$: groups alice

\$: groups bob

\$: groups charlie



The screenshot shows a MobaXterm terminal window titled "13.233.23.129 (ec2-user)". The terminal displays the following commands and output:

```
[root@ip-172-31-12-8 ec2-user]#  
[root@ip-172-31-12-8 ec2-user]#  
[root@ip-172-31-12-8 ec2-user]# sudo usermod -aG devteam alice  
[root@ip-172-31-12-8 ec2-user]# sudo usermod -aG devteam bob  
[root@ip-172-31-12-8 ec2-user]# sudo usermod -aG devteam charlie  
[root@ip-172-31-12-8 ec2-user]#  
[root@ip-172-31-12-8 ec2-user]# groups alice  
alice : alice devteam  
[root@ip-172-31-12-8 ec2-user]# groups bob  
bob : bob devteam  
[root@ip-172-31-12-8 ec2-user]# groups charlie  
charlie : charlie devteam  
[root@ip-172-31-12-8 ec2-user]#
```

The terminal window includes a sidebar with file explorer, a top menu bar, and a status bar at the bottom showing system information and a Windows taskbar overlay.

Create /opt/project_alpha

```
$: sudo mkdir -p /opt/project_alpha
```

Set ownership to: Set ownership to:

```
$: sudo chown root:devteam /opt/project_alpha
```

Set permissions

{ The requirement is:

root → rwx

devteam → rwx

Others → no permissions }

```
$: sudo chmod 770 /opt/project_alpha
```

```
$: ls -ld /opt/project_alpha
```

Enable SetGID

SetGID ensures that files and directories created inside /opt/project_alpha inherit the devteam group.

```
$: sudo chmod 2770 /opt/project_alpha
```

Verify:

```
$: ls -ld /opt/project_alpha
```

✓ Notice the **s** in the group permissions

Test as alice

```
$: su alice
```

Create the required file:

```
$: touch /opt/project_alpha/test_alice.txt
```

Check the file:

```
$: ls -l /opt/project_alpha
```

✓ This proves that the file inherited the devteam group because of the SetGID bit.

Exit from Alice:

```
$: exit
```

Test with a user outside devteam

Create guestuser:

```
$: sudo useradd -m guestuser
```

```
$: sudo passwd guestuser
```

Verify:

```
$: groups guestuser
```

The screenshot shows a MobaXterm terminal window with the following commands and output:

```
charlie : charlie devteam
[root@ip-172-31-12-8 ec2-user]# sudo mkdir -p /opt/project_alpha
[root@ip-172-31-12-8 ec2-user]# sudo chown root:devteam /opt/project_alpha
[root@ip-172-31-12-8 ec2-user]# sudo chmod 770 /opt/project_alpha
[root@ip-172-31-12-8 ec2-user]# ls -ld /opt/project_alpha
drwxrwx---. 2 root devteam 6 Aug 15 13:29 /opt/project_alpha
[root@ip-172-31-12-8 ec2-user]# sudo chmod 2770 /opt/project_alpha
[root@ip-172-31-12-8 ec2-user]# ls -ld /opt/project_alpha
drwxrws---. 2 root devteam 6 Aug 15 13:29 /opt/project_alpha
[root@ip-172-31-12-8 ec2-user]# su - alice
[alice@ip-172-31-12-8 ~]$ touch /opt/project_alpha/test_alice.txt
[alice@ip-172-31-12-8 ~]$ ls -l /opt/project_alpha
total 0
-rw-r--r--. 1 alice devteam 0 Aug 15 13:34 test_alice.txt
[alice@ip-172-31-12-8 ~]$ alice devteam
-bash: alice: command not found
[alice@ip-172-31-12-8 ~]$ exit
logout
[root@ip-172-31-12-8 ec2-user]# sudo useradd -m guestuser
[root@ip-172-31-12-8 ec2-user]# sudo passwd guestuser
passwd: Unknown user name 'guestuser'.
[root@ip-172-31-12-8 ec2-user]# sudo passwd guestuser
```

The terminal window also shows the system status at the bottom: 31°C, Mostly cloudy, 19:06, 15-08-2026.

Switch to guestuser

```
$: su - guestuser
```

Try creating a file:

```
$: touch /opt/project_alpha/test_guest.txt
```

```
touch: cannot touch '/opt/project_alpha/test_guest.txt': Permission denied
```

✓ This proves that users outside devteam cannot write to the directory.

Final verification

```
$: ls -l /opt/project_alpha
```

```
Expected: -rw-r--r-- 1 alice devteam 0 Aug 15 18:00 test_alice.txt
```

The image shows a MobaXterm terminal window with a dark theme. The title bar at the top reads "13.233.23.129 (ec2-user)". The menu bar includes "Terminal", "Sessions", "View", "X server", "Tools", "Games", "Settings", "Macros", and "Help". The toolbar contains icons for Session, Servers, Tools, Games, Sessions, View, Split, MultiExec, Tunneling, Packages, Settings, and Help. On the right, there are buttons for "X server" and "Exit". The left sidebar shows a "Quick connect..." field and a file explorer for "/home/ec2-user/" with files like .ssh, .bashrc, .bash_profile, and .bash_logout. The main terminal area shows a shell session on a root@ip-172-31-12-8 ec2-user machine. The commands and output are as follows:
[root@ip-172-31-12-8 ec2-user]# sudo useradd -m guestuser
[root@ip-172-31-12-8 ec2-user]# sudo passwd gustuser
passwd: Unknown user name 'gustuser'.
[root@ip-172-31-12-8 ec2-user]# sudo passwd guestuser
Changing password for user guestuser.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-12-8 ec2-user]#
[root@ip-172-31-12-8 ec2-user]#
[root@ip-172-31-12-8 ec2-user]# groups guestuser
guestuser : guestuser
[root@ip-172-31-12-8 ec2-user]# su - guestuser
[guestuser@ip-172-31-12-8 ~]\$ touch /opt/project_alpha/test_guest.txt
touch: cannot touch '/opt/project_alpha/test_guest.txt': Permission denied
[guestuser@ip-172-31-12-8 ~]\$
[guestuser@ip-172-31-12-8 ~]\$ exit
logout
[root@ip-172-31-12-8 ec2-user]# ls -ld /opt/project_alpha
drwxrws---. 2 root devteam 28 Aug 15 13:34 /opt/project_alpha
[root@ip-172-31-12-8 ec2-user]# ls -l /opt/project_alpha
total 0
-rw-r--r--. 1 alice devteam 0 Aug 15 13:34 test_alice.txt
[root@ip-172-31-12-8 ec2-user]#
[root@ip-172-31-12-8 ec2-user]#
[root@ip-172-31-12-8 ec2-user]#
[root@ip-172-31-12-8 ec2-user]#
[root@ip-172-31-12-8 ec2-user]#
[root@ip-172-31-12-8 ec2-user]#
[root@ip-172-31-12-8 ec2-user]#
The bottom status bar of the terminal shows "Follow terminal folder" and a list of system metrics: ip-172-31-12-8.ap-south-1.compute.internal, 0%, 0.40 GB / 7.60 GB, 0.01 Mb/s, 0.00 Mb/s, 46 min, ec2-user (x2), /: 12%, /tmp: 0%, /bx. At the very bottom, a banner reads "UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net". The Windows taskbar at the bottom shows the date and time as 19:16 on 15-08-2026, along with system icons for temperature, search, and various applications.